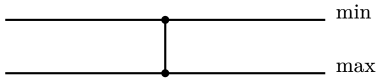
The network djbsort's AVX2 sort runs, at 64 lines

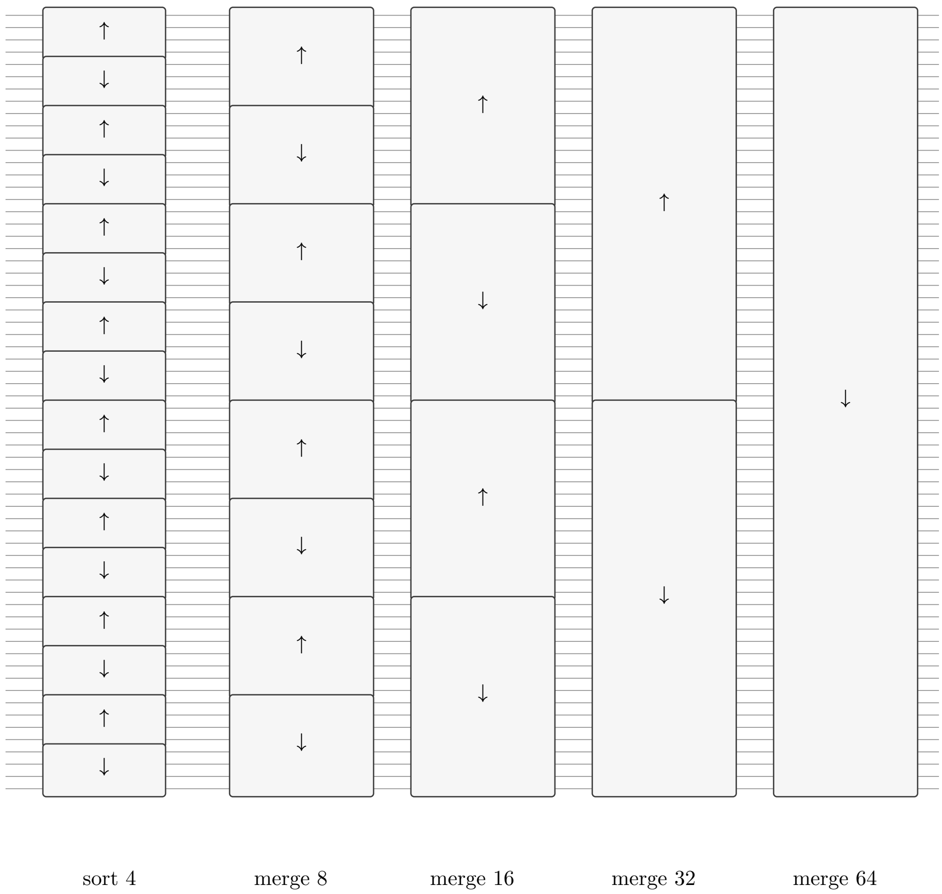
Laurent Théry

A connector joins two lines; the minimum leaves on the upper one.



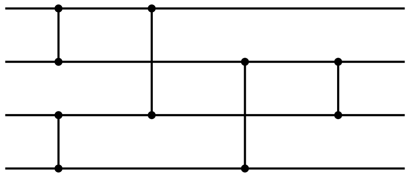
At 64 lines the sort performs 656 comparisons, in 82 groups of eight done at once by one vector instruction. Drawn one comparison at a time the picture is unreadable, so each box below stands for a whole component, and its arrow points towards the larger values.

The shape is a bitonic sort. First every group of four lines is sorted, the upper four decreasing and the lower four increasing, so every group of eight falls and then rises. Then merges of doubling size, each turning two neighbouring runs of opposite direction into one.

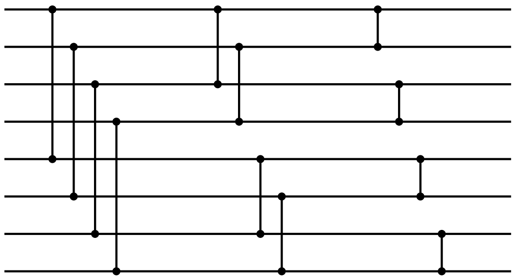


What the boxes contain

sort 4 is the five-comparison network on four lines, at distances $1, 1, 2, 2, 1$. A bitonic sort of four would use six, so this is where the code is smaller than a plain bitonic sorter.



merge 2*m* takes a run down followed by a run up and returns one sorted run. It is the usual cascade: compare at distance m, then $m/2$, down to 1. At 64 lines the merges use distances $4, 2, 1$ then $8, 4, 2, 1$ then $16, 8, 4, 2, 1$ then $32, 16, 8, 4, 2, 1$ — which is what the code performs, once its comparisons are named by the array position each value ends in.



Where the eight-at-a-time grouping sits

A vector instruction compares eight pairs at once. Those eight pairs are one line of the picture repeated across eight places: at every distance the code compares position i with position $i + d$ for eight values of i at once. So the grouping cuts across the boxes rather than following them, and each group of eight is a connector in its own right, its comparisons being on distinct lines.